

MINSK, Belarus

WMO: 268500

Lat: 53.9287N

Lon: 27.6337E

Elev: 734

StdP: 14.31

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	3.0	-8.4	3.6	-1.7	-2.4	5.0	4.2	22.3	33.1	19.5	29.1	3.6	120	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	85.3	68.0	82.2	66.1	78.9	64.7	69.9	81.0	67.9	78.4	66.3	76.3	7.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	98.8	75.4	64.3	92.6	72.8	62.5	87.0	71.2	34.2	81.7	32.7	78.6	31.4	76.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.4	16.2	14.1	DB	-10.1	89.7	7.5	3.3	-15.6	92.1	-20.0	94.0	-24.2	95.9	-29.6	98.3	(4)
(5)			WB	-10.6	72.8	7.5	1.8	-16.0	74.0	-20.4	75.1	-24.6	76.1	-30.0	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.0	23.6	25.1	33.0	46.0	55.9	62.4	66.9	64.8	54.9	43.9	35.5	27.2	(6)
(7)		DBStd	17.24	10.38	10.23	8.34	7.94	7.79	6.17	5.59	6.03	6.37	7.56	7.78	10.06	(7)
(8)		HDD50	3629	819	697	528	168	30	1	0	0	27	216	435	708	(8)
(9)		HDD65	7525	1284	1117	992	572	298	123	44	78	307	653	884	1173	(9)
(10)		CDD50	1817	0	0	1	47	213	371	524	458	174	28	1	0	(10)
(11)		CDD65	236	0	0	0	0	16	43	103	71	3	0	0	0	(11)
(12)		CDH74	2019	0	0	0	10	151	337	855	628	38	0	0	0	(12)
(13)		CDH80	491	0	0	0	1	23	57	220	186	4	0	0	0	(13)
(14)	Wind	WSAvg	6.6	7.0	6.9	7.2	6.8	6.8	6.6	6.0	5.8	6.0	6.4	6.9	7.2	(14)
(15)	Precipitation	PrecAvg	26.7	1.7	1.4	1.6	1.7	2.5	3.1	3.7	2.8	2.2	2.0	2.0	1.9	(15)
(16)		PrecMax	37.5	3.6	3.2	3.7	3.8	5.4	7.4	11.4	10.0	6.1	5.6	4.1	3.8	(16)
(17)		PrecMin	19.8	0.3	0.2	0.3	0.1	0.5	0.5	0.4	0.1	0.5	0.1	0.3	0.6	(17)
(18)		PrecStd	3.9	0.8	0.6	0.7	0.8	1.3	1.6	2.0	1.8	1.2	1.2	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	44.8	58.8	75.1	82.7	85.7	89.3	91.2	78.6	66.5	54.0	46.6	(19)
(20)			MCWB	42.4	41.1	47.6	57.5	64.5	68.1	70.0	68.7	61.7	57.7	51.1	45.2	(20)
(21)		2%	DB	39.4	41.0	53.2	68.8	78.6	82.0	85.8	84.5	73.4	61.0	50.8	43.2	(21)
(22)			MCWB	37.9	38.7	44.3	54.0	63.1	66.2	68.5	67.6	60.2	54.8	48.2	42.1	(22)
(23)		5%	DB	37.4	38.8	48.1	64.4	73.7	78.5	82.3	80.8	69.0	57.5	47.9	40.9	(23)
(24)			MCWB	36.0	37.0	41.3	51.7	60.3	64.6	66.6	65.2	58.9	53.1	46.2	39.9	(24)
(25)		10%	DB	35.3	36.3	44.2	60.1	69.9	74.9	78.9	76.9	65.8	54.9	45.4	37.9	(25)
(26)			MCWB	34.2	34.7	39.5	48.6	57.8	62.6	65.2	63.9	57.2	51.5	43.8	36.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	42.1	49.0	58.5	67.3	70.8	73.0	71.7	64.6	59.7	52.0	45.5	(27)
(28)			MCD B	43.3	43.9	56.5	72.2	79.3	80.9	84.9	84.9	73.6	64.8	53.9	46.3	(28)
(29)		2%	WB	38.2	39.0	45.0	55.3	64.7	68.1	70.3	69.1	61.7	56.5	48.7	42.4	(29)
(30)			MCD B	39.0	40.1	50.5	66.2	75.6	77.9	81.3	80.7	69.6	59.6	50.3	42.9	(30)
(31)		5%	WB	36.0	36.8	42.2	52.9	62.2	66.0	68.3	67.2	59.9	53.8	46.0	39.7	(31)
(32)			MCD B	36.9	38.0	46.6	61.5	71.7	75.4	78.9	77.5	66.8	57.0	47.3	40.7	(32)
(33)		10%	WB	34.0	34.8	39.7	50.5	59.6	63.9	66.6	65.4	58.2	51.6	43.8	36.8	(33)
(34)			MCD B	34.8	36.0	43.3	57.5	67.8	72.4	76.5	74.7	64.1	54.7	45.0	37.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	9.4	12.8	17.8	18.3	17.4	17.8	18.4	16.1	11.9	7.1	7.0	(35)
(36)			MCDBR	6.6	8.8	18.7	24.9	23.3	22.3	22.7	24.5	22.0	16.3	9.6	7.1	(36)
(37)			MCWBR	6.4	7.4	11.5	12.9	11.1	10.6	9.8	10.8	11.7	11.1	7.8	6.6	(37)
(38)		5% WB	MCD BR	6.3	8.1	15.9	21.3	21.1	19.7	20.2	20.2	18.8	14.4	8.4	6.9	(38)
(39)			MCWBR	6.3	7.2	11.0	11.8	11.3	11.1	9.7	9.9	11.2	10.7	7.4	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.288	0.322	0.354	0.409	0.400	0.385	0.415	0.416	0.378	0.351	0.330	0.290	(40)	
(41)		taud	2.166	2.122	2.160	2.187	2.285	2.354	2.282	2.275	2.345	2.399	2.282	2.202	(41)	
(42)		Ebn at Noon	207	237	257	259	269	274	262	254	249	230	191	180	(42)	
(43)		Edh at Noon	21	30	36	41	39	37	39	37	30	23	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	214	471	890	1296	1649	1775	1662	1421	960	482	174	127	(44)	
(45)		RadStd	42	88	137	155	167	142	167	130	88	97	25	34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+1.15	N/A	-342	+272	N/A		
(47)	Regional (11 neighbors)	+0.86	N/A	N/A	N/A	N/A	+1.35	N/A	N/A	N/A	N/A		

Nomenclature: See separate page